AIR-GB Lightning

Multi-path, intent-aware routing for RGB assets

Local AI proposes · deterministic code verifies · Rust routes

35% → 65%

payment success, 20 scenarios

`13/20 vs 7/20 · 81% of reachable`

−79%

fees on payments both completed

`1,200 vs 5,700 msat`

< 1.4 ms

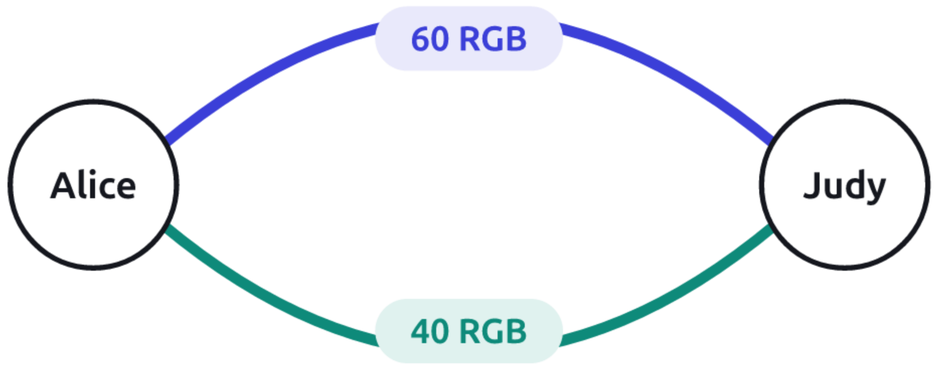
route planning on the node

`0.66–1.35 ms · 4 Rust planners`

Built on rgb-lightning-node · rust-lightning · QVAC local AI

THE OPPORTUNITY

Why multi-path matters as RGB Lightning grows



One path carries at most 60 RGB.

Two paths deliver the full 100.

`max_paths = 1 → max_paths = 2`

1 **Combine liquidity**

Split one payment across parallel channels.

`min-cost flow · 2–8 parts`

2 **Keep channels balanced**

Return flows refill channels, so later payments stay direct.

`reverse MPP · 190 RGB back to Alice`

3 **Follow user intent**

Fee caps, nodes to avoid, privacy, written in plain language.

`max_fee_msat · excluded_nodes · max_paths`

ARCHITECTURE

How a payment gets routed

Baseline stock rgb-lightning-node

7 / 20 succeeded

Payment request → **One shortest path** → **Send**

`POST /sendpayment` · `LDK Dijkstra · max_path_count = 1` · `single HTLC`

AIR-GB

13 / 20 succeeded

1 **Understand intent** — plain-language policy → `interpret.ts → compile.ts`

2 **Recall & score** — local AI + payment memory → `channel-scoring.ts`

3 **Verify & choose** — guardrails pick a strategy → `routing-selector.ts`

4 **Plan in Rust** — multi-path graph planners → `routes.rs · 0.66–1.35 ms`

`Planners: min_cost_flow · multi_objective · yen · ldk_dijkstra`

LOCAL AI

Two small local models, one job each

MEMORY

EMBEDDINGGEMMA_300M_Q4_0

EmbeddingGemma

300M parameters · 4-bit · 768-dim

- Saves every payment outcome
- Recalls the 4 closest matches
- Runs before each payment

`TurboVec (turbovec-q4) + HyperDB rerank`

past payments

top 4 matches

`vector search`

REASONING

QWEN3_5_0_8B_MULTIMODAL_Q4_K_M

Qwen3.5

0.8B parameters · 4-bit · temperature 0

- Reads the routing request
- Scores each channel
- Suggests algorithm and split

`3 passes · strict JSON schema each`

100% on the node · no cloud calls · no keys · only the node's own data

`qvac serve on 127.0.0.1:11434 · first-party episodes only · RAM wiped after each decision`

GUARDRAILS

The AI suggests. Code decides.

Model output → Payment sent

1

Locked output

The model can only name channels that really exist.

`JSON schema enum = live /listchannels IDs`

2

Grounded in words

Fees and nodes must appear in the user's own request.

`secp256k1 key check SHA-256 policy binding`

3

Real balances win

Live balances override the model's channel scores.

`BigInt liquidity floor computed before LLM`

4

Fixed strategy rules

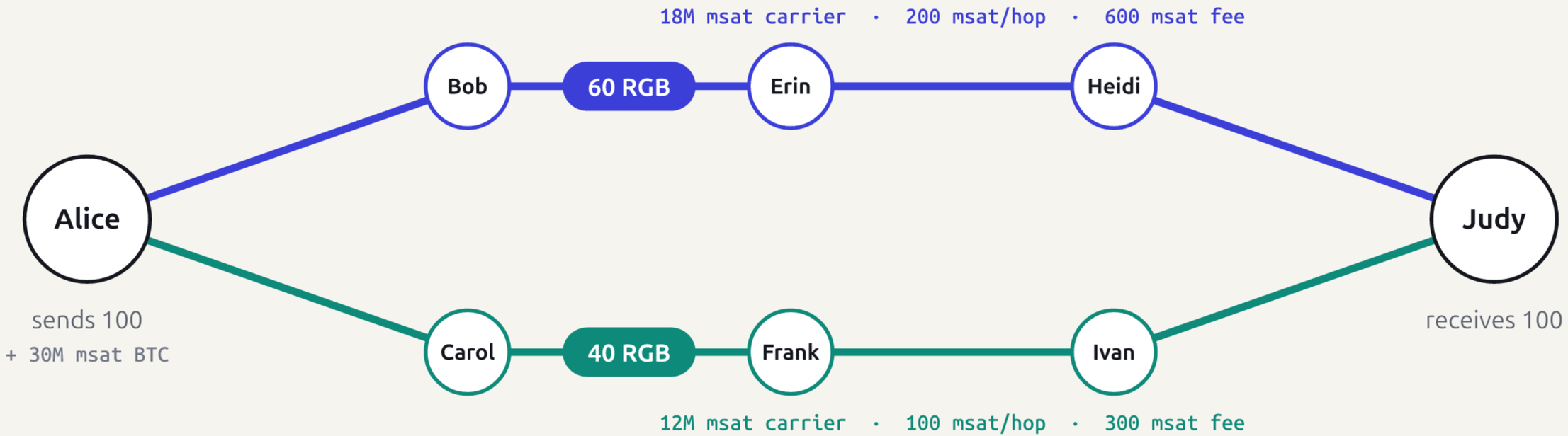
Split when one path can't fit; one path when privacy is asked.

`MPP needed → min_cost_flow privacy → max_paths = 1`

A model mistake is caught before any payment is sent.

MULTI-PATH PAYMENTS

One payment, two paths, settled together



1 Check capacity

Which channels can carry the RGB?

`drop 0-RGB channels · k ∈ [2..8]`

2 Split by cost

Cheaper path first, then the rest

`successive shortest paths`

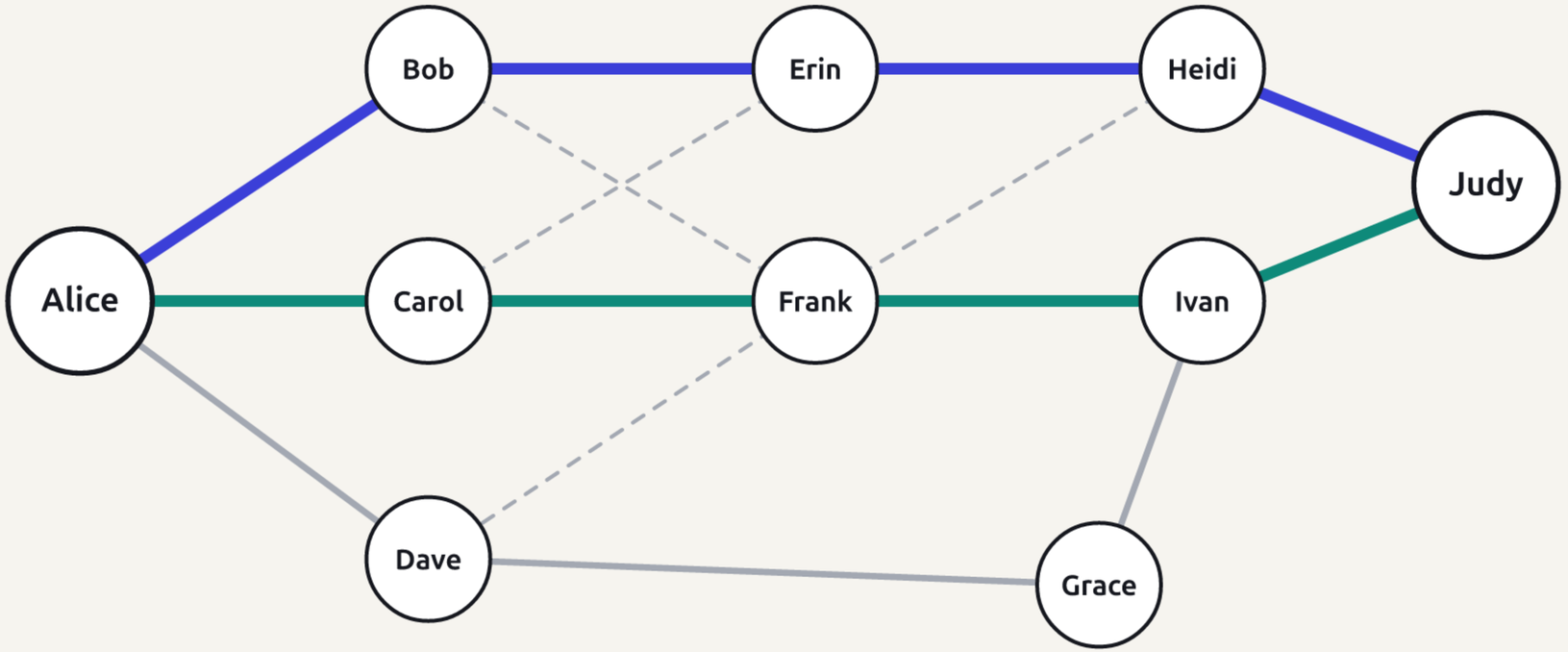
3 Settle atomically

Judy claims only when all 100 arrive

`Σ rgb = 100 ∧ Σ msat ≥ 30M`

BENCHMARK SETUP

Test network: 20 nodes, 15 RGB channels



60 RGB corridor
`200 msat/hop`

40 RGB corridor
`100 msat/hop`

other channels
`0–5,000 msat`

high-fee cross links
`5,000 msat · 20 RGB`

+ 10 isolated nodes as negative controls

Same starting state for both runs
`state hash fc59bc78…8b45`

1M sat channels
`7,000 sat push for returns`

RESULTS

Same network, same 20 payments

Payments completed +30 pts



Fees, payments both completed –79%



WHERE THE GAIN COMES FROM	CASES	BASELINE	AIR-GB
Multi-path payments (60 + 40 RGB)	02 · 03 · 04	0 / 3	3 / 3
RGB returned to the sender	03 · 08 · 10	0	190 RGB
Mid-network payments	12 · 13 · 15	0 / 3	3 / 3
Impossible payments rejected	06 · 17 · 19 · 20	4 / 4	4 / 4

shared successes 01 09 11 14 16 · case 11: 5,100 msat 3-hop detour vs 0 msat direct

LIVE DEMO

From a sentence to a multi-path RGB payment

1

Ask in plain words

“Max 1,000 msat fee,
and avoid Bob.”

`CompiledPolicy + SHA-256 fingerprint`

2

See the AI’s view

Channel scores from
local payment memory

`optimal · good · congested · avoid`

3

Send it

An RGB payment split
across two paths

`min_cost_flow · max_paths = 2`

35% → 65% success · −79% fees · 100% local AI

Thank you. Questions welcome.